

Student Performance Analytics Dashboard

Mega Central High School

Business Analyst Portfolio Project

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Agenda

1. Project Overview
2. Business Problem
3. Project Objectives
4. Scope
5. Stakeholders
6. Proposed Solution
7. Technology Stack
8. Solution Architecture
9. Data Model
10. ETL Process
11. SQL Implementation
12. Power BI Dashboards

Project Overview

- **Project Name: Student Performance Analytics & Decision Support System**
- **Industry: Education (K–12)**
- **Organization: Mega Central High School**
- **Duration: 4 Weeks**
- **Project Type: Business Intelligence & Analytics**

Business Problem

Current Challenges

- Academic data stored in multiple Excel files
- Manual report preparation
- Time-consuming performance analysis
- Delayed identification of struggling students
- Limited curriculum insights
- No centralized reporting system

Business Impact

- Slow decision-making
- Missed intervention opportunities
- Inconsistent reporting
- Increased manual effort

Project Objectives:

The project aims to:

- Centralize student assessment data
- Automate performance reporting
- Monitor academic progress
- Identify at-risk students
- Analyze subject performance
- Identify weak chapters
- Support data-driven academic decisions

Project Scope

In Scope

- Student Data
- Test Data
- Question Bank
- Student Responses
- Python ETL
- SQL Database
- Power BI Dashboards
- Documentation
- Testing

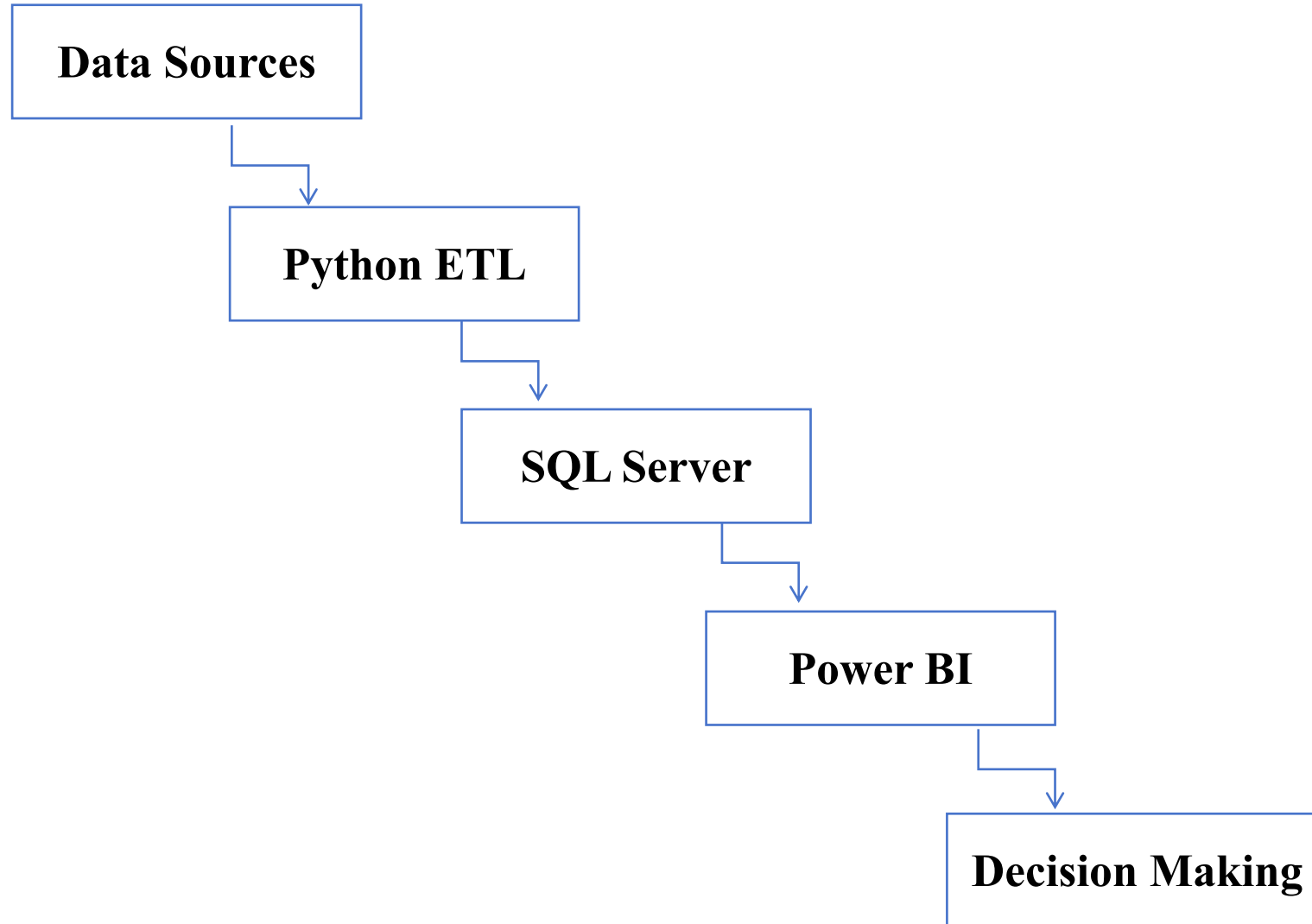
Out of Scope

- Student Information System Integration
- Parent Portal
- Mobile Application
- Machine Learning
- Real-time Data

Stakeholders:

Stakeholder	Role
Principal	Strategic Decision Making
Academic Coordinator	Curriculum Monitoring
Teachers	Student Performance Tracking
Students	Academic Improvement
Parents (Future)	Progress Monitoring

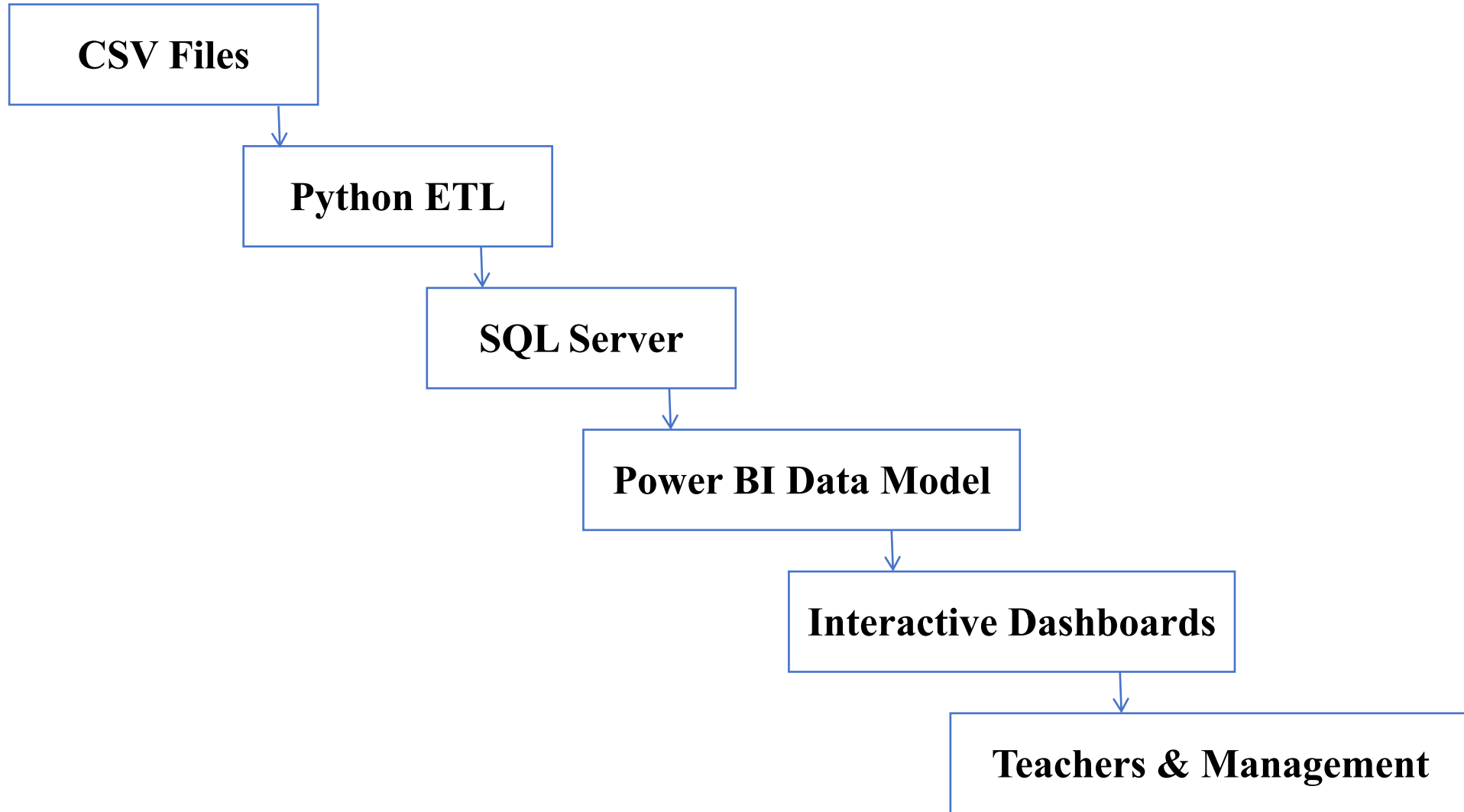
Proposed Solution



Technology Stack

Layer	Technology
Programming	Python
Data Processing	Pandas
Database	Microsoft SQL Server
Query Language	SQL
Visualization	Power BI
Version Control	GitHub

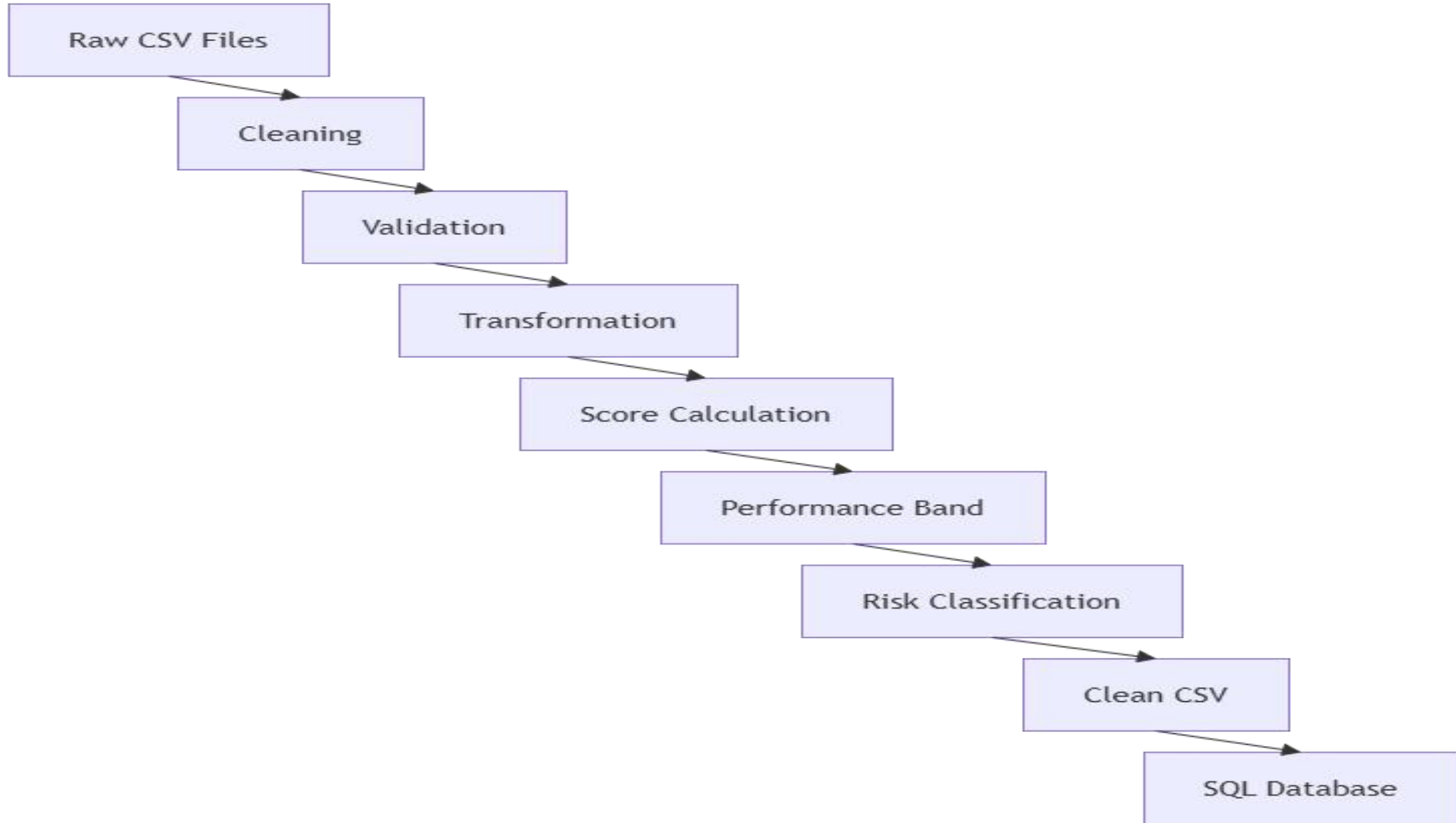
Solution Architecture



Data Model



ETL Pipeline



SQL Implementation

Database:

StudentPerformanceAnalytics

Tables:

- students
- tests
- responses
- question_bank
- chapter_analytics
- question_analytics
- scores
- student_progress
- student_risk_report

SQL Server acts as the centralized repository for cleaned educational data and supports business reporting through validation and analytical queries.

Examples:

Top Students

High Risk Students

Subject Performance

Weak Chapters

Power BI Dashboard - Executive Performance Overview

Student Performance Analytics Dashboard Mega Central High School

Total Students
2000

Tests Conducted
50

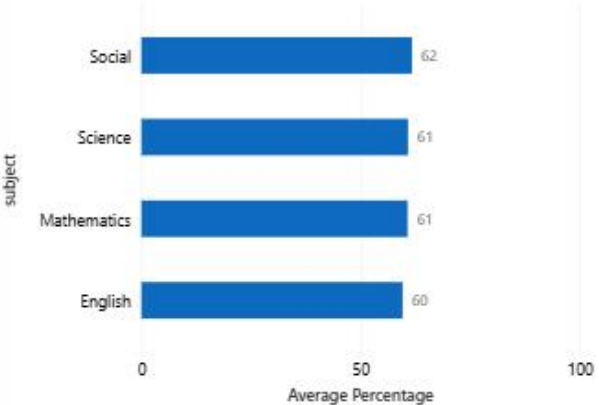
Average Score
60.76%

Pass Percentage
85.87%

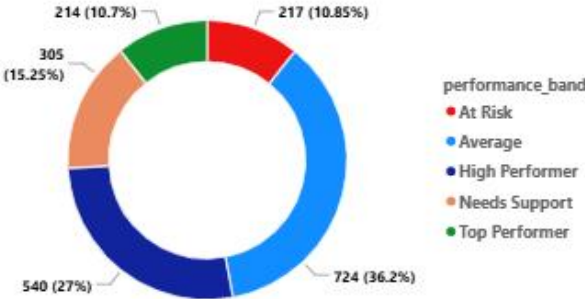
High Risk Students
286

Top Performers
214

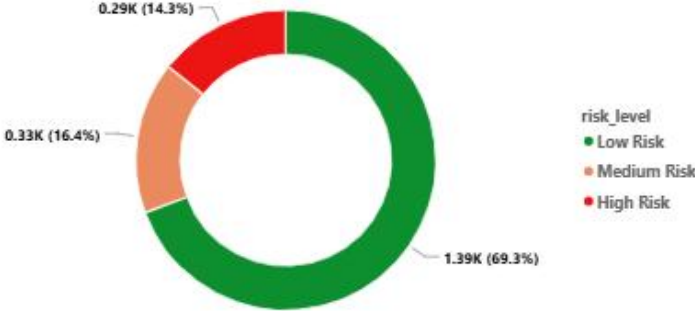
Subject Performance



Student Performance Distribution



Overall Student Risk Profile



Monthly Academic Performance Trend



Power BI Dashboard - Individual Student Performance Analysis

Student Performance Analytics Dashboard Mega Central High School

Select Student

Utkarsh Grover

Student Name

Utkarsh Grover

Student Grade

10

Student Risk Level

High Risk

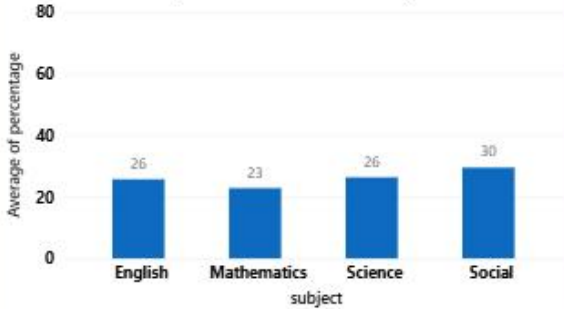
Student Performance Band

At Risk

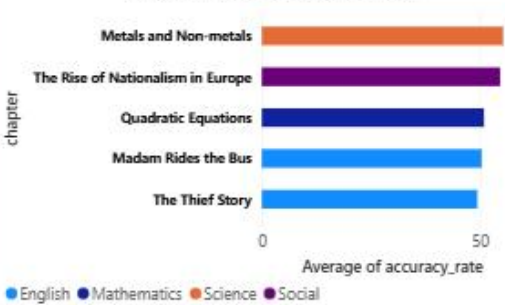
Student Average Score

25.81

Subject Performance Analysis



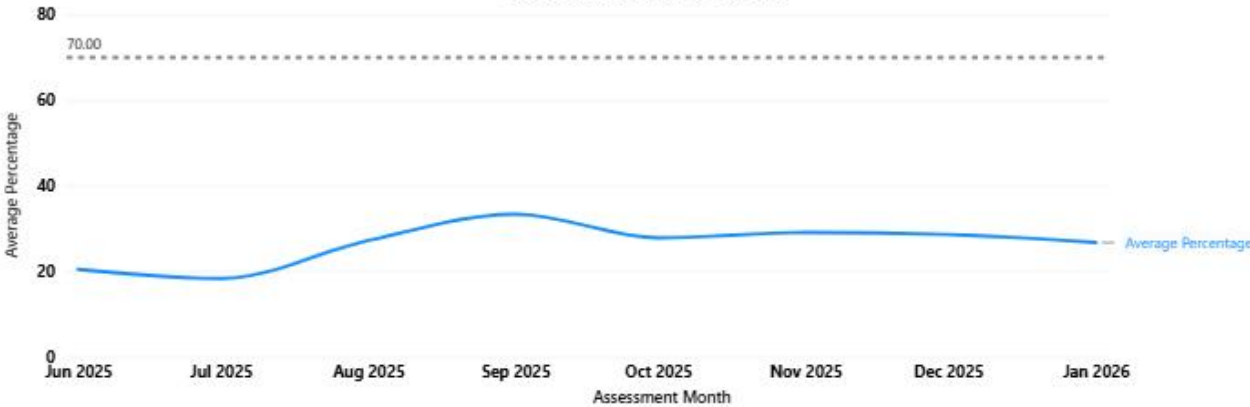
Chapter-wise Learning Gaps



Assessment History

Test ID	date	Subject	Test Type	Score Percentage	Grade	Pass / Fail
T002	11-06-2025	Mathematics	Unit Test	25.96	E	Fail
T003	12-06-2025	Mathematics	Unit Test	25.96	E	Fail
T006	19-06-2025	Social	Unit Test	25.96	E	Fail
T014	01-08-2025	Science	Unit Test	25.96	E	Fail
T019	25-08-2025	English	Unit Test	25.96	E	Fail
T020	26-08-2025	Science	Unit Test	25.96	D	Pass
T023	30-08-2025	English	Unit Test	25.96	E	Fail
T024	18-09-2025	Social	Unit Test	25.96	E	Fail
T025	08-10-2025	Social	Unit Test	25.96	D	Pass
T038	20-11-2025	Science	Unit Test	25.96	E	Fail
T013	23-07-2025	Science	Revision Test	25.96	E	Fail
T026	09-10-2025	English	Revision Test	25.96	C2	Pass
T028	19-10-2025	Mathematics	Revision Test	25.96	E	Fail
T032	03-11-2025	English	Revision Test	25.96	E	Fail
T035	10-11-2025	Science	Revision Test	25.96	E	Fail
T004	14-06-2025	Science	Pre-Board	25.96	E	Fail
T007	25-06-2025	Mathematics	Pre-Board	25.96	E	Fail
T044	29-12-2025	Science	Pre-Board	25.96	E	Fail
T005	14-06-2025	Science	Monthly Test	25.96	E	Fail
T011	17-07-2025	English	Monthly Test	25.96	E	Fail
T012	22-07-2025	Mathematics	Monthly Test	25.96	E	Fail
T018	23-08-2025	Mathematics	Monthly Test	25.96	E	Fail
T021	27-08-2025	Social	Monthly Test	25.96	E	Fail
T029	27-10-2025	Mathematics	Monthly Test	25.96	E	Fail
Total				25.96		

Student Performance Trend



Power BI Dashboard - Risk Analysis Dashboard

Student Performance Analytics Dashboard Mega Central High School

Total Students
2000

High Risk Students
286

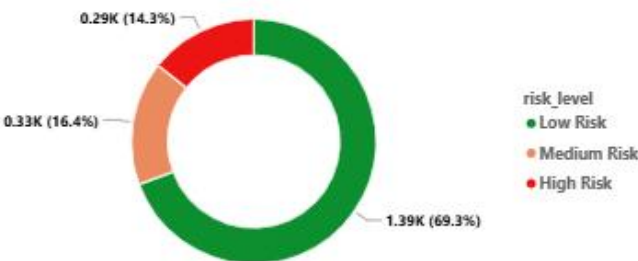
Medium Risk Students
328

Low Risk Students
1386

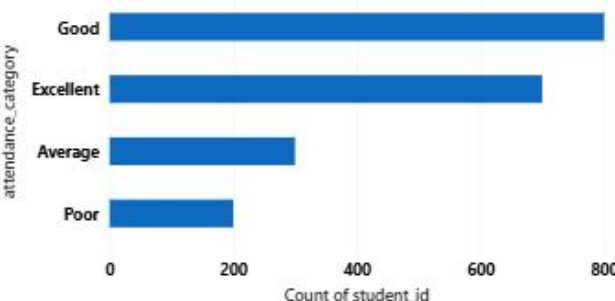
Average Score
60.76%

Students Needing
Academic Support
614

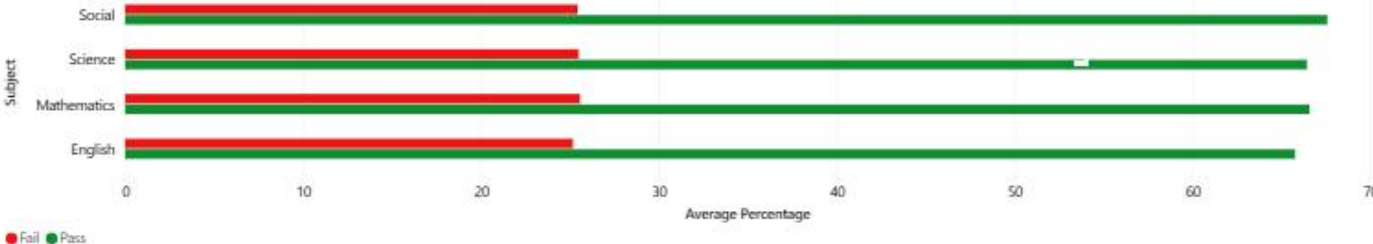
Overall Student Risk Profile



Attendance Category Distribution



Subject-wise Risk Assessment



High Risk Students List

Student name	Grade	Risk Level	Average Percentage	Attendance Category
Aanya Tiwari	10	High Risk	22.32	Average
Abha Malhotra	10	High Risk	29.40	Excellent
Abha Prabhu	10	High Risk	23.33	Good
Abha Pradhan	10	High Risk	25.69	Poor
Aditi Bhat	10	High Risk	17.49	Poor
Aditya Bose	10	High Risk	25.43	Excellent
Aditya Joshi	10	High Risk	26.55	Average
Aditya Mukherjee	10	High Risk	26.75	Average
Akash Menon	10	High Risk	34.37	Good
Akash Roy	10	High Risk	29.09	Excellent
Alka Iyer	10	High Risk	22.55	Poor
Alka Reddy	10	High Risk	34.21	Good
Alok Naik	10	High Risk	27.18	Excellent
Amit Das	10	High Risk	25.33	Poor
Amita Hegde	10	High Risk	26.56	Excellent
Ananya Singh	10	High Risk	17.16	Average
Aniruddh Grover	10	High Risk	27.38	Excellent
Aniruddh Jain	10	High Risk	30.12	Excellent
Anita Shetty	10	High Risk	35.42	Good
Anjali Acharya	10	High Risk	26.39	Excellent
Anjali Gowda	10	High Risk	34.45	Average
Total	2170		6,048.29	

Power BI Dashboard - Subject & Chapter Performance Analysis

Student Performance Analytics Dashboard Mega Central High School

Total Subjects
4

Total Chapters
55

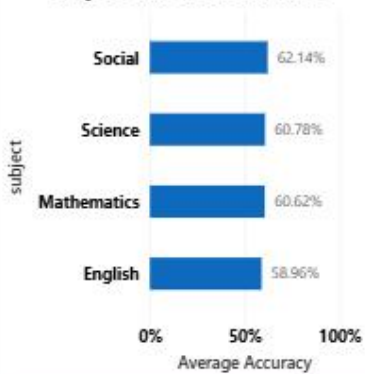
Average Accuracy
60.62%

Chapters Below Target
4

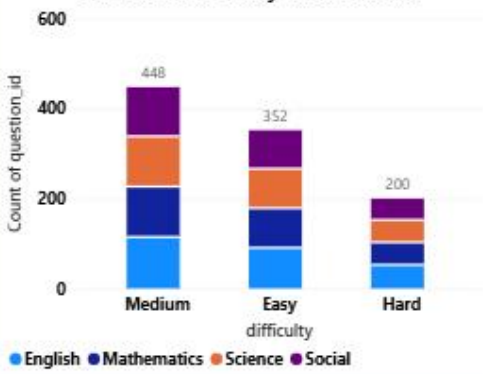
Chapters Requiring Attention

chapter	subject	Average of accuracy_rate
A Letter to God	English	67.78
A Question of Trust	English	55.66
A Triumph of Surgery	English	63.01
Acids, Bases and Salts	Science	64.20
Agriculture	Social	65.01
Areas Related to Circles	Mathematics	63.25
Arithmetic Progressions	Mathematics	61.75
Carbon and its Compounds	Science	60.33
Chemical Reactions and Equations	Science	61.90
Circles	Mathematics	62.39
Control and Coordination	Science	58.87
Coordinate Geometry	Mathematics	62.78
Development	Social	59.04
Electricity	Science	57.35
Federalism	Social	61.34
Footprints without Feet	English	58.19
Forest and Wildlife Resources	Social	69.49
From the Diary of Anne Frank	English	60.74
Gender, Religion and Caste	Social	65.11
Glimpses of India	English	61.07
Heredity	Science	63.50
How do Organisms Reproduce?	Science	61.64
Introduction to Trigonometry	Mathematics	65.59
Life Processes	Science	55.73
Light - Reflection and Refraction	Science	61.88
Madam Rides the Bus	English	50.42
Magnetic Effects of Electric Current	Science	63.85
Metals and Non-metals	Science	55.35
Total		60.62

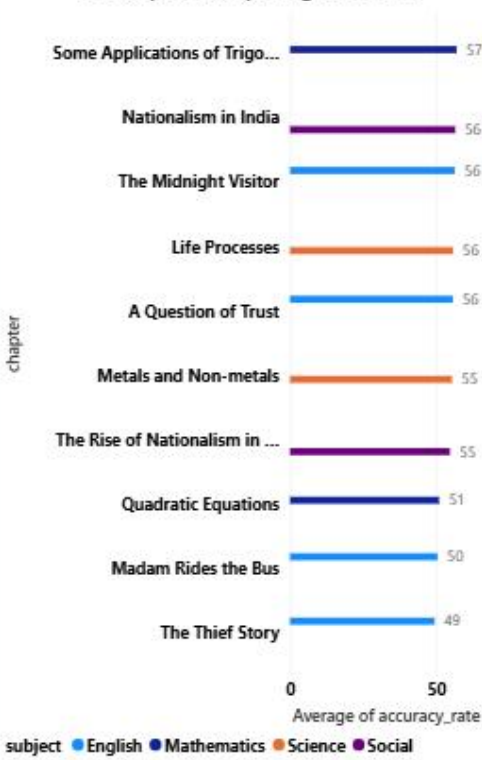
Subject-wise Performance



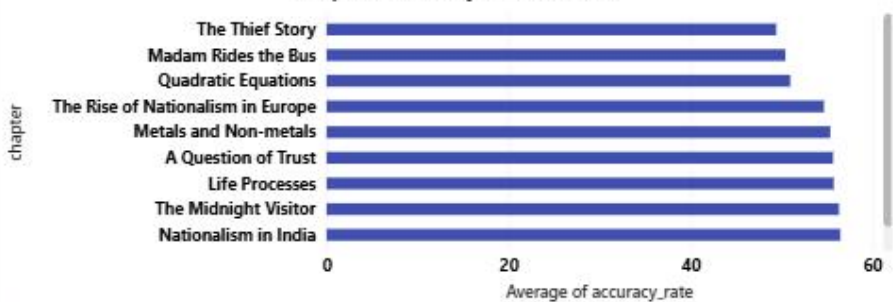
Question Difficulty Distribution



10 Chapters Requiring Attention



Chapter Accuracy Distribution



Dashboard Highlights

Executive Dashboard:

- KPIs
- Monthly Trend
- Subject Performance

Student Dashboard:

- Individual Performance
- Weak Chapters
- Assessment History

Risk Dashboard:

- High-Risk Students
- Attendance Analysis
- Subject-wise Risk

Subject Dashboard:

- Subject Comparison
- Question Difficulty
- Chapter Accuracy

Business Insights

- Centralized view of school-wide academic performance through interactive KPIs.
- Early identification of high-risk students enables timely academic intervention.
- Subject-wise comparisons reveal relative strengths and areas requiring improvement.
- Chapter-level analysis identifies learning gaps to support targeted remedial teaching.
- Student performance trends help monitor academic progress across assessments.
- Question difficulty analysis assists in evaluating the quality and balance of assessments.
- Interactive dashboards reduce manual reporting and improve decision-making efficiency.
- Supports data-driven planning for teachers, academic coordinators, and school leadership.

Testing Summary

Documents Created:

- Test Plan
- Test Cases
- Defect Log
- UAT Scenarios
- UAT Sign-off

Testing Completed:

- Functional Testing
- Data Validation
- UAT

Challenges & Solutions

Challenge	Solution
Poor Dataset	Redesigned Dataset
SQL Import Errors	Increased Column Sizes
Duplicate Data	Python ETL Validation
Relationship Issues	Redesigned Data Model
KPI Accuracy	Validated with SQL & DAX

Business Benefits

Operational Benefits:

- Automated Reporting
- Centralized Database
- Reduced Manual Effort

Academic Benefits:

- Student Risk Identification
- Curriculum Analysis
- Personalized Intervention

Strategic Benefits:

- Data-Driven Decision Making
- Improved Academic Planning

Future Enhancements

- Parent Portal
- Teacher Dashboard
- Predictive Analytics
- AI Recommendations
- Real-Time SQL Integration
- Attendance Analytics
- Mobile Dashboard
- Cloud Deployment

Key Learnings

Business Analysis:

- Requirement Gathering
- Stakeholder Analysis
- Documentation
- Process Mapping

Technical skills:

- Python ETL
- SQL Server
- SQL Queries
- Data Validation
- Power BI
- DAX
- Dashboard Design

Soft Skills:

- Problem Solving
- Analytical Thinking
- Documentation
- Communication

Conclusion

Project Outcome:

The Student Performance Analytics & Decision Support System successfully transformed fragmented assessment data into a centralized Business Intelligence solution. By integrating Python for ETL, Microsoft SQL Server for data storage and business querying, and Power BI for visualization, the project enables school stakeholders to monitor academic performance, identify at-risk students, evaluate curriculum effectiveness, and make data-driven decisions.

This project demonstrates the complete Business Analyst lifecycle—from requirement gathering and documentation to ETL development, SQL implementation, dashboard creation, testing, and stakeholder-ready reporting.

Thank You

Questions & Discussion